

Chocolate Enterprises Report

Analyzing the Chocolate Enterprises Report.

A) Evaluate this report in detail following the Visual Analytics methodology for the report phase.

Answer: In first analysis, the profit trends over time and quantity sorted by age groups are also taken into account, and it is successfully takes the profit made by each sales channel catalog, internet, and retail store. A clear and effective understanding of the important metrics is made possible by the visualization tools that are employed, with button bar functionality, switching between channels and viewing the appropriate client data dynamically is simple. The revenue swings over the last four years are clearly broken down in the profit by order date bar chart, which also highlights growth trends and performance variances. The most important financial statistic is always in plain sight thanks to the key value meter, which prominently displays the overall profit. The distribution of order quantities among age groups is skillfully depicted in the pie chart, making it easy to see which demographic makes the biggest and smallest contributions to sales. Because of the pie chart's proportional sizing and color coding, the analysis is visually important. The layout and chart selection provides gurantee that information is presented without any mistake. The color scheme maintain a clean and captivating style while making the facts easy to understand.

The analysis second portion, which focuses on retail performance, highlights the top-selling items, sales volume, and profits made. Additionally, the top ten retail establishments with the largest sales are listed. When combined, a tree map and a bar chart offer an engaging and informative summary of sales performance. When a retail store is selected, the tree map is updated to show product-specific information for that particular location. Double-clicking enables customers to examine sub-product information and go down into particular product categories. The arrangement of the bar chart should be improved because certain viewers could find it challenging to understand when profit and total sales volume are shown on the same axis. Clarity might be improved by using a different kind of graphic, which would guarantee that the profit and sales statistics are easily distinguished.

B) On the Customer Analysis page, identify and describe any actions, prompts, and links used on it, and provide step-by-step instructions on how to create them in SAS Viya.

Answer:

Step 1: Create a Page in the Customer Enterprises report to do the analysis part and identify each values.

Step 2: Next, add the name “Customer Analysis”, click “page 1” twice or by clicking the “three points” on the right side of that particular title bar, and then “rename”.

Step 3: Create dashboards by taking three channels of data, which allows for the analysis of each part separately. For doing that, I have to create a 'Button bar' which will consist of the horizontal bar representation that gives a clear picture, next click on the object on the left of the interface, then scroll down till the controls section and select the Button bar. Click on the assign data in the page and click on the category for the assigned 'Order Channel' data. In turn, it displays the button bar with three channels that need to be analyzed.

Step 4: Moving forward, include three variety charts(Profit by order, Item quantity, Total profit). For inclusion of the channel total profit, select 'Objects' on the left side of the panel. Choosing the chart 'Key value'. Providing profit to measure, which depicts the revenue in words or text, to modifying it to a visualization, requires selecting the option on right side of the panel, then clicking on the style to choose the graph.

Step 5: In order to display the item amount by age group in visualization, select the objects click on the Pie chart option, add the age group to the class item quantity to measure and verify. Next, click the settings, navigate to the labels of data section, and verify the box next to Total percent of values to depict the percentage.

Step 6: Additionally, once more select the objects and click on the bar chart by assigning Order date steps to category, then click Profit to calculate, to include the bar chart that displays profit by order date. Next, to modify the title, select 'Profit by date' steps from the options on the right side, then click on the object name. The name profit by order date is included and the title is changed to custom. The title can be edited by selecting it twice on the bar chart and adjusting the font, color, and alignment of the text.

Step 7: The unique button at the top of each chart can be dragged to move the charts. This allows for easy change of the position of the chart to align.

Step 8: Finally, as a last step, connect each order channel to its unique information. The exact statistics for this channel should be depicted if it's clicked on the order channel button. Next, go to the activities on the left side of the page after clicking the button bar. Moreover, select automation actions for each action.

C] Examine the rest of the Customer Analysis page. Identify all the objects presented on this page, except from the prompts or actions identified in I.(b). Describe how they are created and explain why they are included, which requires you to describe SAS report methodology (8 guidelines).

Answer:

In the Catalog Sales channel, this dashboard is providing examination of purchasing patterns, client demographics, and earnings. With 2020 being the most profitable year and producing \$126,268.18, the total profit stands at \$462K, 2018 and 2019 followed with \$114,837.12 and \$111,580.13, respectively. Over the years, the profit has increased steadily, with 2017 yielding the lowest profits at \$109,549.82. Customers aged 30-44 make the most purchases (40.5%),

followed by 19-29 years (24.2%) and 45-59 years (19.4%), according to the pie chart on Quantity Ordered by Age Group. This suggests that the majority of consumers are middle-aged individuals. Younger and older groups, however, make comparatively smaller contributions. In order to determine where people like to shop, the dashboard also compares sales across the various channels (Catalog, Internet, and Retail Stores). By understanding client behavior over time and identifying important trends are made easier by the visual representation provided by bar charts and pie charts.

Next, Internet Sales channel, the overall profit of \$994K, which is the subject of this Customer Analysis dashboard on Internet sales, is much higher than the catalog sales (\$462K) in the earlier figure. 2018 saw the largest profit of \$262,833.14, which was followed by 2017 with \$253,627.00. Profits for the remaining two years, 2020 and 2019, were \$243,449.37 and \$234,146.92, respectively, displaying consistent profit over time. The most group of buyers (37.6%) are in the 30-44 age range, followed by the 19-29 age group (25.1%) and the 45-59 age group (22.1%), according to the quantity ordered by age group pie chart. Unexpectedly, only 0.2% of purchases were made by consumers 75 years of age or older, indicating that older consumers are not very involved. In addition to highlighting the significance of focusing on youth and old individuals for online sales, the comparison of profit by year and client graphs also points to possible development opportunities among other age groups.

The Retail Store channel, The profit trend over time is among the most important features. Profit increased constantly between the years 2017 and 2020, as seen in the bar chart titled Profit by Order Date. The profit increased to \$7.96M in 2020 from \$3.34M in 2017. With increasing income each year, this rising trend indicates that the company has been expanding consistently. The section on quantity ordered by age group provides another important information. 5.1 million orders were placed, according to the donut chart, and a significant 46.1% of those orders originated from a certain age group and providing that a specific graphs is driving sales, it is a useful target market for upcoming marketing and sales initiatives. Overall, the research shows a better growth in business, mainly in retail shop sales, and shows how important it is to understand customer graphs.

Improving the Chocolate Enterprises Report.

DJ Prepare to further analyze the data set. Before starting your analysis for the tasks below, you should access and investigate the data set, including understanding its structure, data types, variables, dimensions, and possible data issues. Provide clear instructions on how you access and explore the data set, and write a detailed description of the data set.

Answer:

First, log into SAS Viya is necessary in order to access the data set and examine the data in the software. Then click Show List of Applications in the top left corner, and choose Visualize and Interactive. SAS Visual Analytics emerges. Select “all reports”, then “building_reports/activity_4_final” under “Courses/GAVA85”. The chocolate enterprise report can be accessed there.

The list of data items in the Category group can be viewed by selecting Data in the left pane. Numerous sorts of information pertaining to the customer's purchase are included in the data collection, including numeric variables (age, dates, profit, item amount) and character variables (customer_name, Order Channel, Item Package type). There are more than 40 categories, and the data kinds are displayed here. The data items in the measure group are visible if we scroll down in the Data window. If we consider and observe the data set, we can see the following information:

- a) Customers information: This customers information consists of various data sets like, Customer name, Customer age, Date of Birth, Card program member or not, Customer self description, City name and Bought information
- b) Order Information: This information provides the various kind of data that includes, Order type, Order channel, Order date, Post code, Transaction line, Item price, Order of Quantity, Transaction Id, Profit and Marginal profit.

E] Produce a Location Analysis report page in SAS Viya to provide a summary of the global and regional profits for Chocolate Enterprise company's senior management. These facts should include financial, marketing related data, and/or efficient use of resources. Capture a screenshot of this page after you have created it, and include it in the coursework report. Describe how you have chosen the appropriate visualization tools. How did you choose the objects in your report (charts, tables, etc.)? What questions would they ask and what answers could they derive based on your report?

Answer:

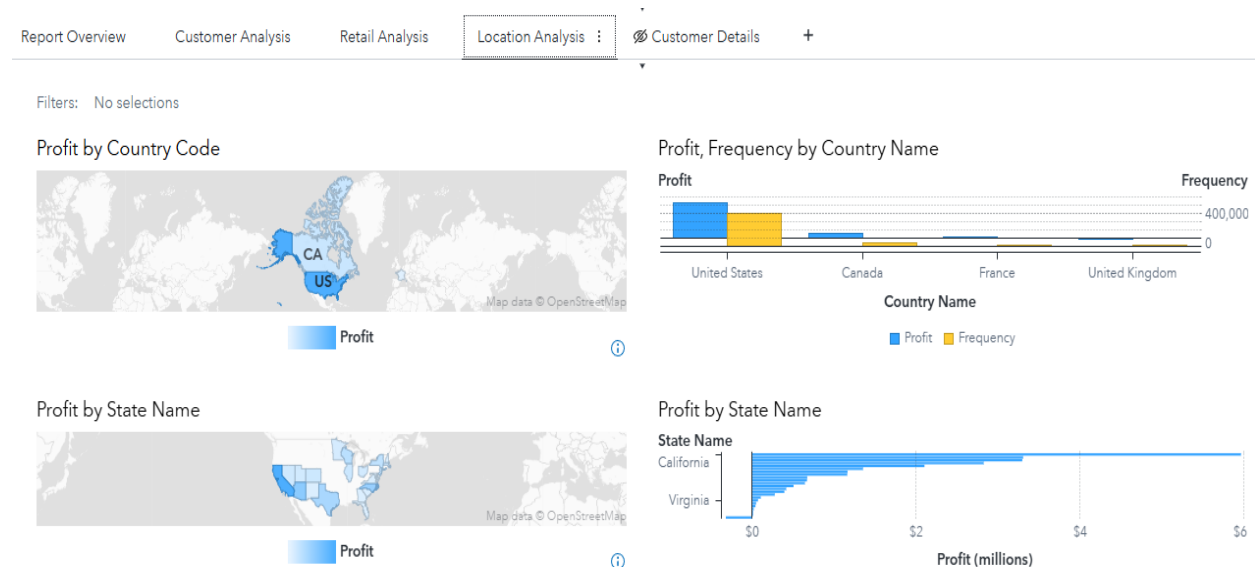


Figure 1: Overview of Location Analysis

The above figure represents a Location Analysis dashboard in SAS viya, which provides insights into profit distribution by country and state and also the dashboard consists of different key visualizations that display profit trends across various locations.

In the first top left image we can see that map chart visualizes the profit distribution across different countries using the light blue color. The dark color of blue represent higher profits, while light indicates low profit. The United states and Canada are highlighted, indicating that these countries have recorded profits in the dataset.

Next, we can see the Top right representing the bar chart that compares profit and frequency of transactions across different countries, X-axis displays different countries(U.S, Canada, France, U.K) and Y-axis represents the profit and transaction frequency where, U.S has highest profit and transaction frequency with Canada as moderate profit and transaction frequency and France, U.K shows lesser value in both metrics.

Followed by, in the Bottom left we can see the mapchart provides breakdown of profit by state within the U.S. Similarly to country level map, states are in light blue and dark blue displaying higher profits. There are specific states that are contributing to profits can be identified based on intensity of color.

Finally, to the Bottom right, there is bar chart representing more detailed comparison of profit by state. With X-axis providing the amount of profit generated by each state and Y-axis displays the state names. With, California has the highest record of profit and Virginia with some notable amount of profit and others have smaller levels.

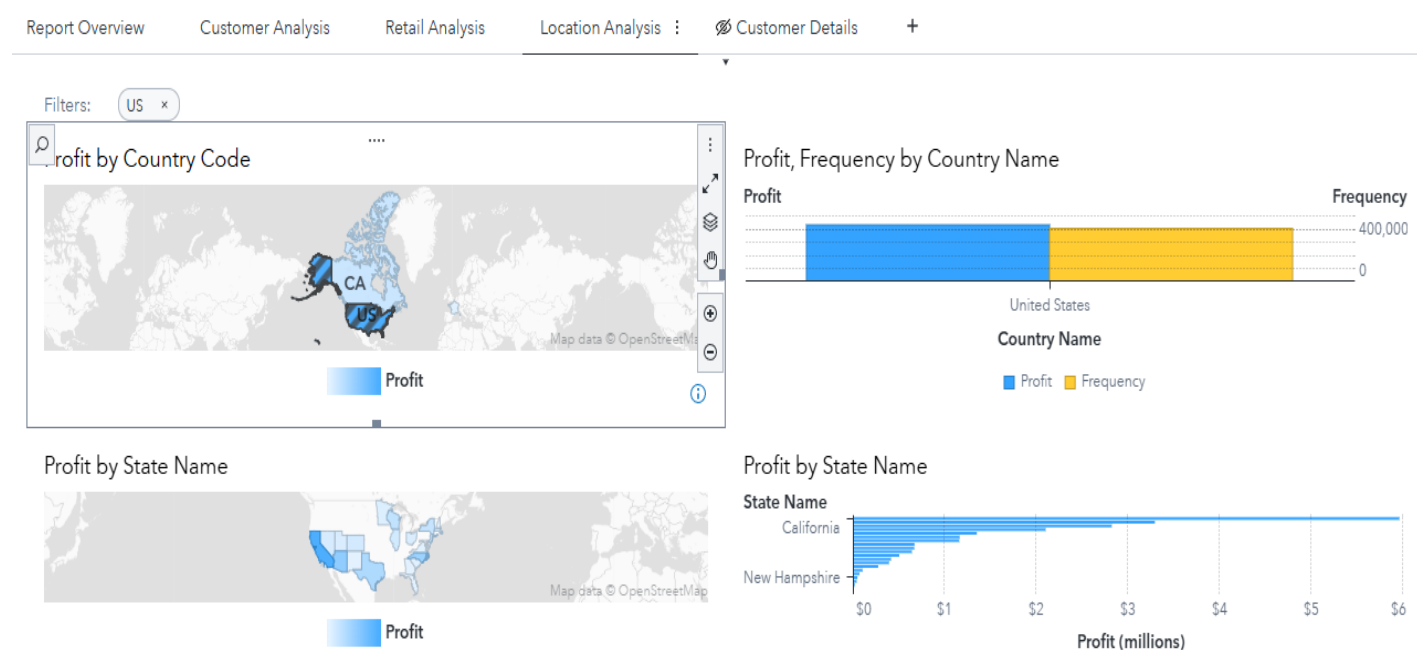


Figure 2: Profit by Country Code

Here, in the above figure if we are selecting the Country code as U.S , the profit, frequency of that particular is changed according to their country,the profit with \$21,781,574.66 and frequency of 408,844.

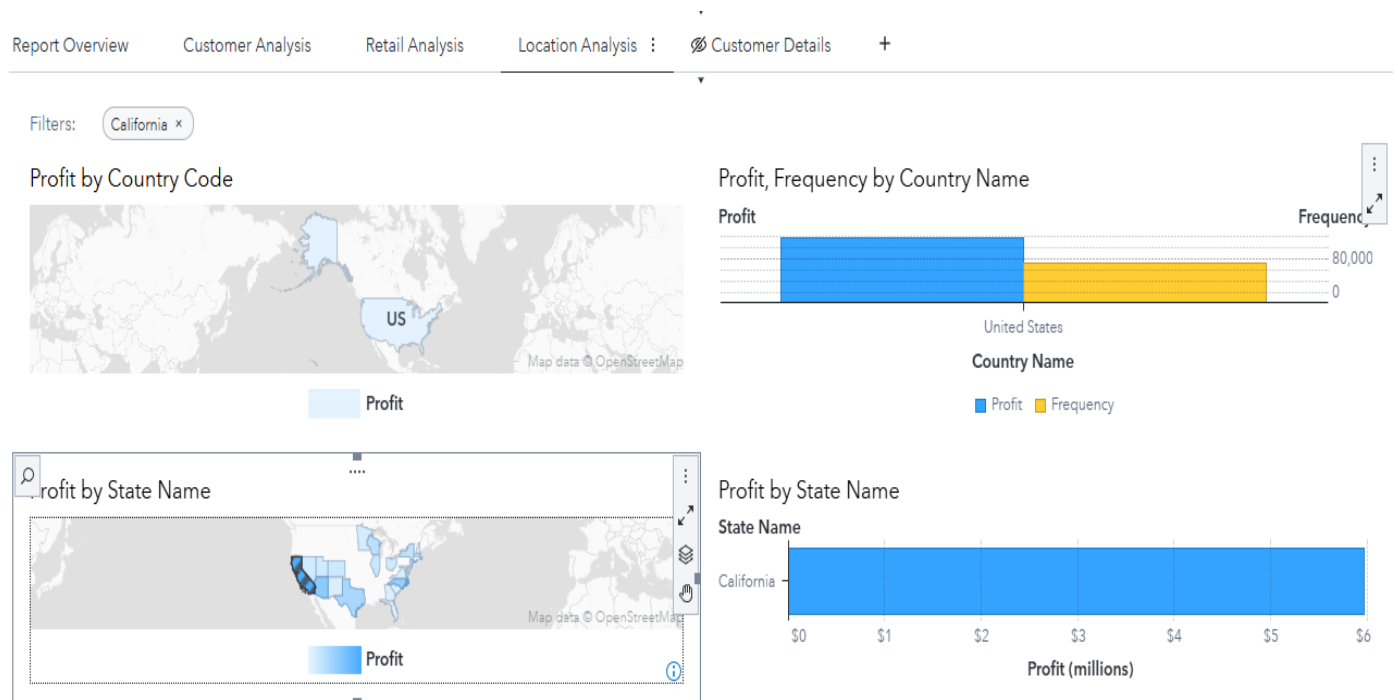


Figure 3: Overview of particular State(California)

Similarly, I have selected the state in the U.S (California) when I click on that state it is displays the above image with showing the profit of \$5,970,872.80 and with frequency of 73.573.

Bonus task: Produce an Overview page as the first page that summarizes all the three pages, using a variety of interactions such as links to connect with other report pages. You can refer to the Orion Star Sports and Outdoors report.

Answer:

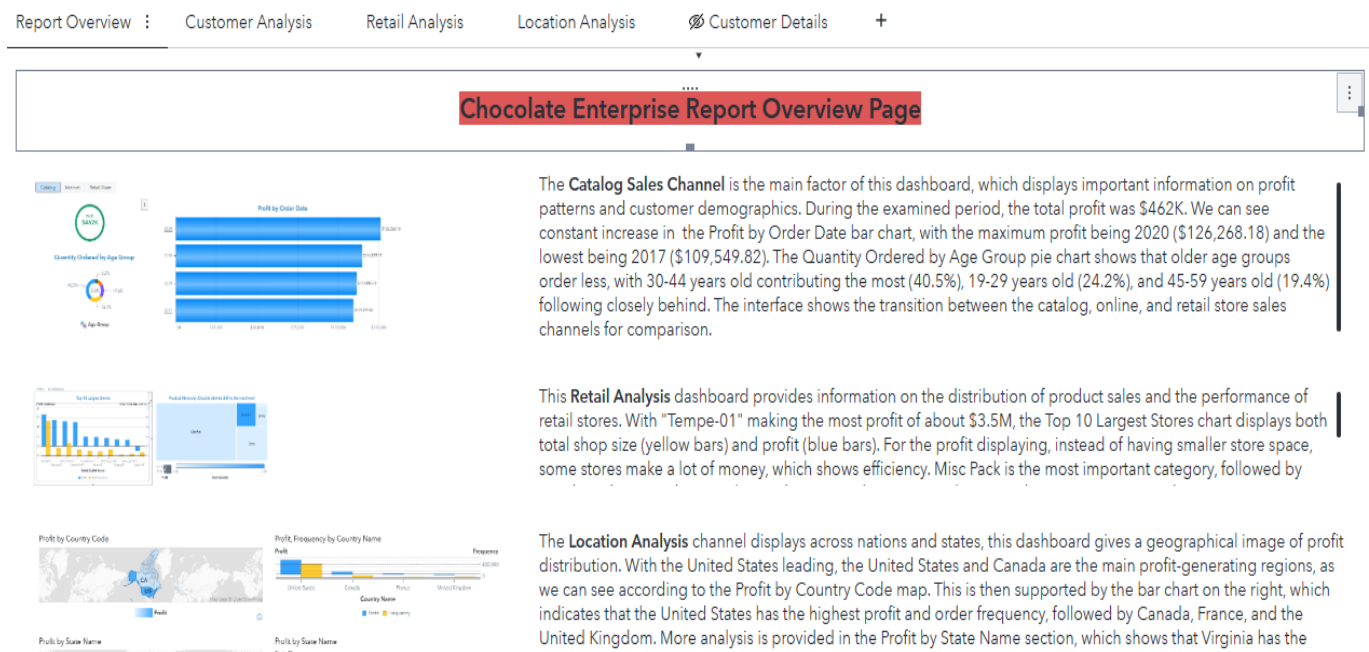


Figure 4: Chocolate Enterprise Report Overview Page

The Chocolate Enterprise Report Overview provides a deeper overview of the company's performance in several domains. We can see the largest source of profit is coming from the Catalog Sales Channel, which has grown stable over the years, with 2020 being the most profitable. Then the customers who are older, particularly those between the ages of 30 and 44, typically place larger orders. The Retail Analysis explains that how some stores generate large profits in their smaller spaces, with some product categories performing better than others. Virginia is the highest earning state, as we can see according to the Location Analysis, which also shows that the United States and Canada generate the most revenue. To make the comparison easier we can take various sales channels, client categories, and regions, the report makes use of clear charts and maps.

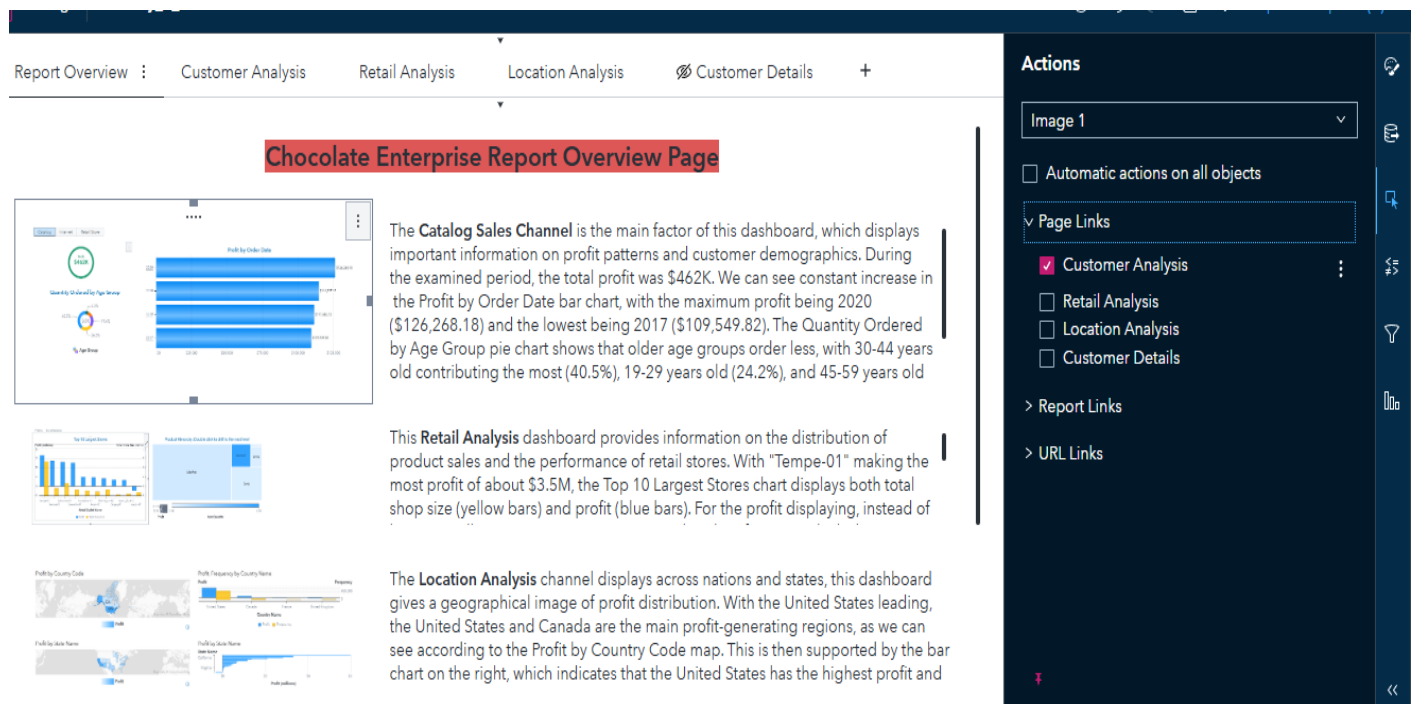


Figure 5(IMAGE 1) : Page link for created image to Customer Analysis

For making the page link, I clicked on the three dots and then selected the “Add link” option in that by clicking “Page link” selected the “Customer Analysis” for the above image and it has been linked successfully and we can see the link proof on the right side of the interface.

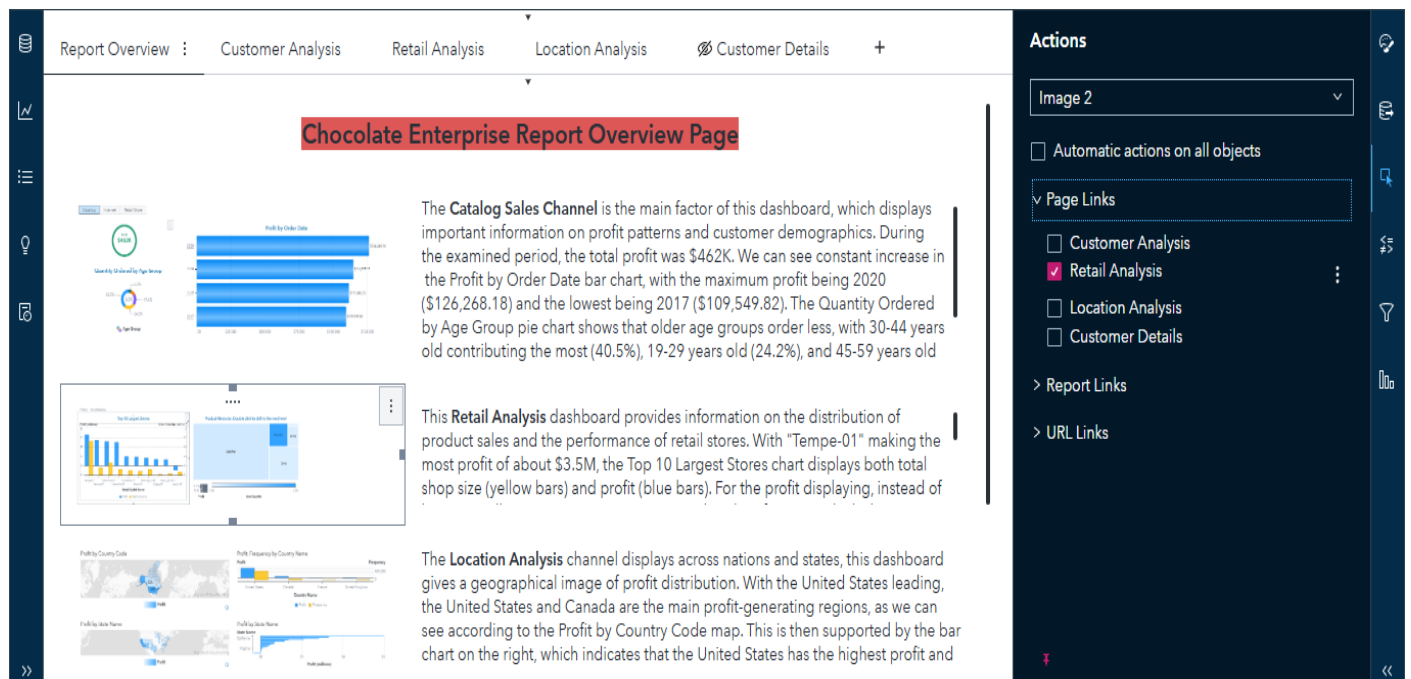


Figure 6(IMAGE 2): Page link for created image to Retail Analysis

Similarly, for the above image page link, I clicked on the three dots and then selected the “Add link” option in that by clicking “Page link” selected the “Retail Analysis” and it has been linked successfully and we can see the link proof on the right side of the interface.

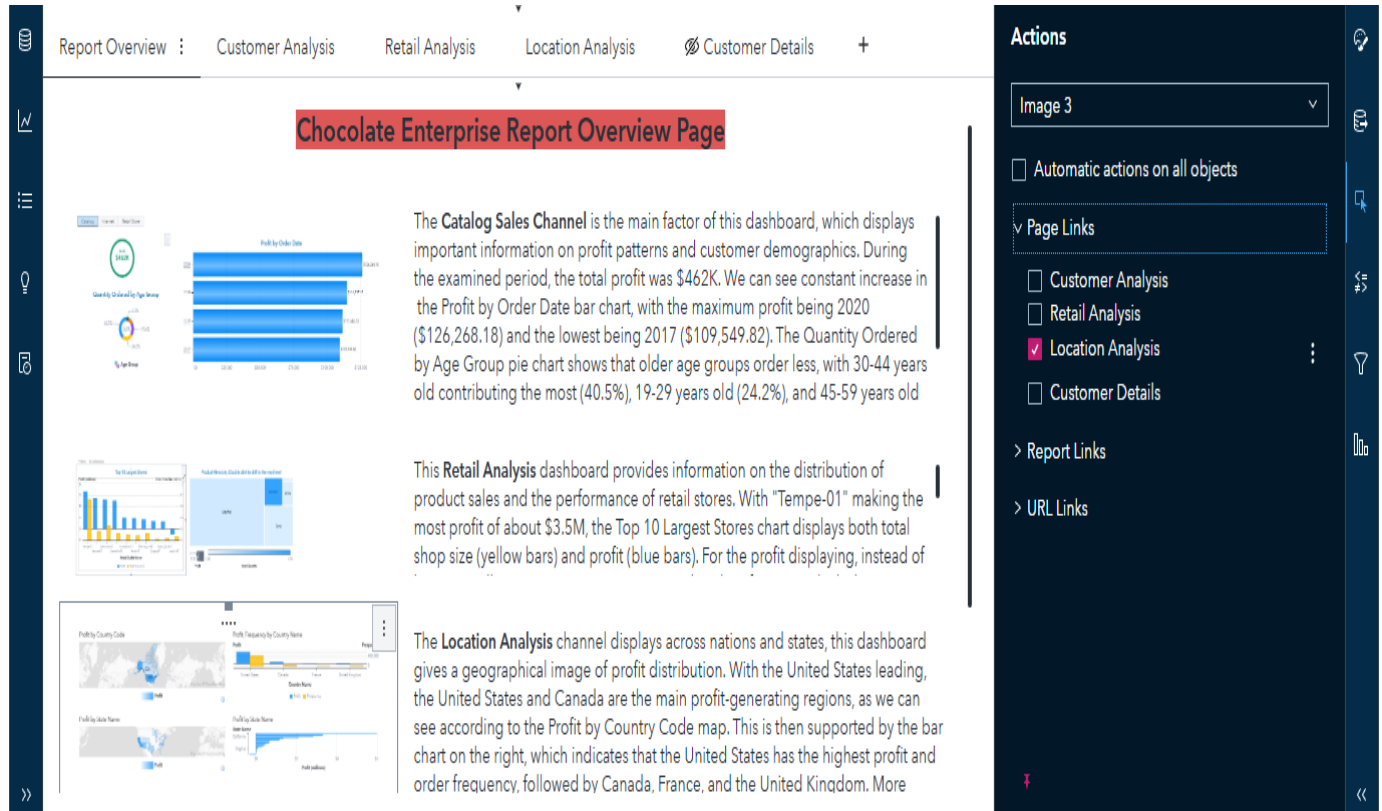


Figure 7(IMAGE 3): Page link for created image to Location Analysis

Finally like the above steps, I clicked on the three dots and then selected the “Add link” option in that by clicking “Page link” selected the “Location Analysis” and it has been linked successfully and we can see the link proof on the right side of the interface.